SET current and motion

From capacity differences to gravitational response

Erik Echeverria
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A burden gradient has a definite role in SET motion. It enters the probe's energy and momentum through the same capacity relation that describes its event progression. Here we follow that connection from finite burden differences to radial motion and the spherical orbit.

$$S^2 + |\mathbf{V}|^2 + c^2\beta^2 = c^2, \qquad \alpha^2 = 1 - \frac{S^2}{c^2}, \qquad \beta^2 = \alpha^2 - \frac{|\mathbf{V}|^2}{c^2}. \tag{1}$$

Here S is gravitational burden expressed in speed units, $\mathbf{V}$ is the specified physical displacement reading per reference time T, and $\beta = d\tau/dT$ is the probe's event progression. A held clock has rate α. Work on the massive-probe branch $\beta > 0$, with a fixed rest mass m. Define $U = S^2/2$.

To compare two locations, specify a spatial sampling coordinate χ, increasing from point 1 to point 2. Its finite current reading is

$$J_{12} \equiv \frac{S_1^2 - S_2^2}{2\Delta\chi}, \qquad \Delta\chi = \chi_2 - \chi_1 > 0. \tag{2}$$

This has acceleration units. Subtracting the capacity relation at the two locations gives exactly

$$J_{12} = \frac{|\mathbf{V}_2|^2 - |\mathbf{V}_1|^2 + c^2(\beta_2^2 - \beta_1^2)}{2\Delta\chi}. \tag{3}$$

The current reads capacity differences, with its sign recording the direction of change. This definition introduces neither a separate transported substance nor a trajectory or conservation law.

For held readings, both velocities vanish and $\beta_i = \alpha_i$. With the clock-normalized burden $q_i = S_i/\alpha_i$, the same finite difference becomes

$$J_{12} = \frac{c^2(\alpha_2^2 - \alpha_1^2)}{2\Delta\chi} = \alpha_1^2\alpha_2^2\frac{q_1^2 - q_2^2}{2\Delta\chi}. \tag{4}$$

The endpoint factors preserve the burden difference across the clock normalization. Lowercase q has speed units; it is distinct from the source quantity Q.

For a smooth field, $J_\chi = -\partial U/\partial\chi$. Its numerical value depends on the specified physical distance convention for χ.

The connection to energy uses the probe coupling already retained in SET. Take the action $I_p = \int L_p \, dT$, with $L_p = -mc^2\beta$, and define the positive event reading $W = mc^2\beta$. Then

$$\mathbf{P} = -\frac{\partial W}{\partial \mathbf{V}} = \frac{m\mathbf{V}}{\beta}, \qquad E = W + \mathbf{P}\cdot\mathbf{V} = \frac{mc^2\alpha^2}{\beta}. \tag{5}$$

At fixed m, differentiating the capacity relation and then W gives

$$d\beta = -\frac{dU + \mathbf{V}\cdot d\mathbf{V}}{c^2\beta}, \qquad dW = -\frac{m}{\beta}\,dU - \mathbf{P}\cdot d\mathbf{V}. \tag{6}$$

Differentiate $E = W + \mathbf{P}\cdot\mathbf{V}$. The terms containing $d\mathbf{V}$ cancel, leaving

$$dE = \mathbf{V}\cdot d\mathbf{P} - \frac{m}{\beta}\,dU = \mathbf{V}\cdot d\mathbf{P} + \frac{m}{\beta}J_\chi\, d\chi. \tag{7}$$

The last form applies to a burden variation along χ. It is an exact differential identity. Energy conservation along a physical path will follow from the stationary probe action below; it has not been inserted into equation (7).

The coefficient of the current also appears when momentum is differentiated with respect to velocity at fixed burden:

$$\mathcal{M}_{ij} = \frac{m}{\beta}\left(\delta_{ij} + \frac{V_iV_j}{c^2\beta^2}\right), \qquad \mathcal{M}_\perp = \frac{m}{\beta}, \qquad \mathcal{M}_\parallel = \frac{m\alpha^2}{\beta^3}. \tag{8}$$

The same expression fixes the current's energy coefficient and the probe's inertial response. This is the connection developed in *SET inertia* [1]. The two coefficients are not separately adjusted.

For radial motion, let ℓ be the physical path coordinate and write $V = d\ell/dT$, with its sign retained. Introduce the rapidity θ by

$$V = c\alpha\tanh\theta, \qquad \beta = \alpha\,\mathrm{sech}\,\theta, \qquad u \equiv \frac{V}{\alpha} = c\tanh\theta. \tag{9}$$

These expressions substitute directly into the invariant. The local held-frame speed is u; θ is a useful velocity parameter. Momentum and reference energy simplify to

$$P = mc\sinh\theta, \qquad E = mc^2\alpha\cosh\theta. \tag{10}$$

[1] Erik Echeverria, SET inertia: Inertia and the response of finite field capacity (2026).

The current acquires a direct motion reading through the same action. For a stationary radial field, distinguish the physical-distance gradient from the work-current:

$$J_\ell = -\frac{dU}{d\ell}, \qquad d\chi = \alpha\, d\ell, \qquad J_{\text{work}} = -\frac{dU}{d\chi} = \frac{J_\ell}{\alpha} = c^2\frac{d\alpha}{d\ell}. \tag{11}$$

The final equality follows by differentiating $U = c^2(1 - \alpha^2)/2$. The conversion $d\chi = \alpha\, d\ell$ defines the work sampling coordinate once the physical distance ℓ is supplied. It does not determine ℓ from the sources.

Because $\partial L_p/\partial\ell = -(m/\beta)J_\ell$, the free-probe Euler-Lagrange equation is

$$\frac{dP}{dT} + \frac{m}{\beta}J_\ell = 0. \tag{12}$$

Now use $P = mc \sinh\theta$ and $\beta = \alpha/\cosh\theta$. Before imposing equation (12), its left-hand side is exactly

$$\frac{dP}{dT} + \frac{m}{\beta}J_\ell = m\cosh\theta\left(c\frac{d\theta}{dT} + J_{\text{work}}\right). \tag{13}$$

Since $m \cosh\theta$ is positive, the equation of motion becomes

$$c\frac{d\theta}{dT} = -J_{\text{work}}. \tag{14}$$

The work-current is minus c times the rate of change of radial rapidity. If burden decreases in the outward direction, J_{work} is positive and θ decreases: the local outward speed decreases; a bound probe turns and then moves inward. Equation (14) remains regular at $\theta = 0$. No division by the probe's velocity was needed.

The energy balance confirms the same result. Along this stationary path, $dU/dT = -J_\ell V$, so equation (7) gives

$$\frac{dE}{dT} = V\left(\frac{dP}{dT} + \frac{m}{\beta}J_\ell\right) = 0. \tag{15}$$

For successive local held-clock readings, $d\tau_h = \alpha\, dT$. Differentiating $u = c\tanh\theta$ gives

$$\frac{du}{d\tau_h} = -\frac{J_{\text{work}}}{\alpha}\left(1 - \frac{u^2}{c^2}\right). \tag{16}$$

This compares velocities in local held frames along the path. It is not the proper acceleration recorded by an accelerometer carried by the freely falling probe. In the weak-burden, low-speed limit, $\alpha \approx 1$ and $u \ll c$, so the motion reduces to $dV/dT \approx -J_\ell = \partial(S^2/2)/\partial\ell$.

The spherical orbit calculation carries the spatial conversion explicitly. In an equatorial plane, use a stationary radial field and the physical velocity reading

$$|\mathbf{V}|^2 = a(r)^2\dot{r}^2 + r^2\dot{\varphi}^2, \qquad d\ell = a(r)\,dr, \qquad D(r) \equiv \alpha a = \frac{d\chi}{dr} > 0. \tag{17}$$

Dots denote derivatives with respect to T, and r is the circumferential radius. The function $a(r)$ converts radial coordinate intervals to physical length. Set $B = \alpha^2$. Time and angular independence of the same action give the conserved specific readings

$$\varepsilon \equiv \frac{E}{mc^2} = \frac{B}{\beta}, \qquad j \equiv \frac{r^2\dot{\varphi}}{\beta}. \tag{18}$$

For a nonradial orbit, choose the angular orientation so that $j > 0$. Put $w = 1/r$, and use a prime for d/dφ. Then $\beta = B/\varepsilon$ and $\dot{r}/\beta = -jw'$. Substituting these into the invariant gives

$$D^2(w')^2 + Bw^2 = \frac{c^2}{j^2}(\varepsilon^2 - B). \tag{19}$$

Differentiate with respect to φ. Write $B_w = dB/dw$ and $D_w = dD/dw$. Away from a turning point, dividing the common factor $2w'$ yields

$$D^2w'' + DD_w(w')^2 + Bw = -\frac{B_w}{2}\left(w^2 + \frac{c^2}{j^2}\right). \tag{20}$$

The radial Euler-Lagrange equation gives the same expression without dividing by w'. Thus equation (20) also applies at a smooth orbital turning point and to a circular orbit.

The current conversion is fixed by its definition:

$$J_{\text{work}} = \frac{c^2}{2D}\frac{dB}{dr}, \qquad B_w = -\frac{2DJ_{\text{work}}}{c^2w^2}. \tag{21}$$

The orbit equation can therefore be written entirely in burden-current and distance readings:

$$D^2w'' + DD_w(w')^2 + Bw = \frac{DJ_{\text{work}}}{c^2} + \frac{DJ_{\text{work}}}{j^2w^2}. \tag{22}$$

This shows how the radial current enters the angular orbit. It also preserves the distinction between the field profile and the physical distance over which that profile changes. A change in the current's normalization must carry the corresponding distance conversion.

For a spherical source domain of radius R_0 and reference volume $V_\Omega = 4\pi R_0^3/3$, the source law supplies

$$\mu \equiv GM = \frac{Q_\Omega^2}{24\pi V_\Omega} = \frac{Q_\Omega^2}{32\pi^2 R_0^3}. \tag{23}$$

Use its retained exterior field and propagation coupling, with the unloaded reference at infinity:

$$S^2 = \frac{2\mu}{r}, \qquad B = 1 - \frac{2\mu}{c^2 r}, \qquad a = B^{-1/2}, \qquad D = 1, \qquad J_{\text{work}} = \frac{\mu}{r^2}. \tag{24}$$

This branch has $r > 2\mu/c^2$. Because $D = 1$, the work coordinate χ agrees with r up to a constant, although the physical radial length satisfies $d\ell = dr/\sqrt{B}$. The finite-domain source law fixes μ; equation (24) also states the retained exterior response used in the application.

Substitution into equation (22) gives

$$w'' + \left(1 - \frac{2\mu w}{c^2}\right) w = \frac{\mu w^2}{c^2} + \frac{\mu}{j^2}. \tag{25}$$

Moving the burden term on the left to the right gives the spherical orbit equation:

$$w'' + w = \frac{\mu}{j^2} + \frac{3\mu w^2}{c^2}. \tag{26}$$

In equation (25), the burden term on the left contributes two units of $\mu w^2/c^2$, and the current on the right contributes one.

For the leading weak-field advance, let $p = j^2/\mu$ and use the Kepler orbit $w_0 = [1 + e_{\text{orb}} \cos\varphi]/p$. The resonant cosine term in $3\mu w_0^2/c^2$ is $6\mu e_{\text{orb}} \cos\varphi/(c^2 p^2)$. Matching it with the shifted phase $\cos[(1 - \delta)\varphi]$ gives

$$\delta = \frac{3\mu}{pc^2}, \qquad \Delta\varpi \simeq 2\pi\delta = \frac{6\pi GM}{a_{\text{orb}}(1 - e_{\text{orb}}^2)c^2}. \tag{27}$$

Here $p = a_{\text{orb}}(1 - e_{\text{orb}}^2)$ at Kepler order. The result is the retained spherical perihelion advance, now reached through the current's energy and motion relations.

The current connects capacity differences, energy exchange and changing motion within one probe coupling. The spherical application retains its established field and radial measure; these identities give their dynamical consequences without independently selecting them.

Verification: 63 symbolic equalities and 30,000 assertions across 5,000 rational states passed. Nine trajectories in a supplied smooth test field crossed a radial turning point, with maximum relative energy drift 3.4×10^{-13}. These are mathematical and numerical checks, not observations.